Worksheet 1

Q8 - 11

Question 8. Let $\beta \in \mathbb{R}$ be a lower bound for a set $B \subset \mathbb{R}$.

Prove that $\beta = \inf B$ if and only if for all $\varepsilon > 0$,

there exists $b \in B$ such that $b < \beta + \varepsilon$.

Let $L \in \mathbb{R}$ be a lower bound for a set $B \subseteq \mathbb{R}$ if

$$L \leq b \; \forall b \in B \, .$$

The negation of the above, i.e.,

Suppose $x \in \mathbb{R}$ is not a lower bound for a set $B \subseteq \mathbb{R}$.

Then $\exists b \in B \ni x > b \, .$

Proof. $\Rightarrow$) Given $\beta = \inf B$.

Let $\varepsilon > 0$. Then β is the greatest lower bound for B and hence $\beta + \varepsilon$ is not a lower bound for B, so there exists $b \in B$ such that $b < \beta + \varepsilon$.

Take $x = \beta + \varepsilon$ and it is not a lower bound for B.

Then $\exists b \in B \ni x > b$.

$\Leftarrow$) Given that for all $\varepsilon > 0$,

there exists $b \in B$ such that $b < \beta + \varepsilon$.

Suppose $m = \beta + \delta = \inf B$ for some $\delta > 0$.

Then there exists $b \in B$ such that

$$b < \beta + \varepsilon = m$$

(when we choose $\varepsilon = \delta$), a contradiction

(since m is a lower bound for B).

How to prove existence?

How to prove uniqueness?

Question 9. Let $\phi \neq A \subset \mathbb{R}$ be bounded below. Prove that inf A is unique.

Proof. (Existence) Infimum for A is guaranteed by the completeness axiom.

(Uniqueness) If A has only one element, then inf A is *the* element and it is unique.

Suppose there exist two infima for A,

say $\beta, \beta' \in \mathbb{R}$.

Since β is the greatest lower bound for A

and β' is a lower bound, it follows that

$$\beta \leq \beta' .$$

By the similar argument, $\beta' \leq \beta$.

Hence, $\beta' = \beta$.

Question 10. Suppose that $\left|a-L\right|<r$, and $\left|b-L\right|<r$ where $r>0$. Show that

(a) $\left|2a+5b-7L\right|<7r$ (b) $\left|3a-2b-L\right|<5r$

Proof. We use the triangle inequality to prove the following.

(a) $\left|2a+5b-7L\right|=\left|2(a-L)+5(b-L)\right|\leq\left|2(a-L)\right|+\left|5(b-L)\right|<2r+5r=7r$.

(b) $\left|3a-2b-L\right|=\left|3(a-L)-2(b-L)\right|\leq\left|3(a-L)\right|+\left|-2(b-L)\right|=3\left|a-L\right|+2\left|b-L\right|$
$<3r+2r=5r.$

Question 11. Let $r > 0$ and $\delta = \min\{1, \frac{r}{9}\}$.

Prove that $\left|x-4\right| < \delta \Rightarrow \left|x^2-16\right| < r$.

Proof. Note that $\delta = \min\{1, \frac{r}{9}\} = \begin{cases} 1, & 1 \le \frac{r}{9} \\ \frac{r}{9}, & 1 > \frac{r}{9} \end{cases}$.

Case 1: $\delta = 1 \Leftrightarrow 1 \le \frac{r}{9} \Leftrightarrow 9 \le r$. Applying the triangle inequality:

$$\left|x-4\right| < 1 \Rightarrow \left|x+4\right| = \left|x-4+8\right| \le \left|x-4\right| + 8 < 1+8 = 9.$$

$$\Rightarrow \left|x^2-16\right| = \left|x-4\right|\left|x+4\right| < \left|x+4\right| < 9 \le r.$$

Case 2: $\delta = \frac{r}{9} \Leftrightarrow \frac{r}{9} < 1 \Leftrightarrow r < 9$. Applying the triangle inequality again:

$$\left|x-4\right| < \tfrac{r}{9} < 1 \Rightarrow \left|x+4\right| = \left|x-4+8\right| \leq \left|x-4\right| + 8 < 9\,.$$

$$\Rightarrow \left|x^2-16\right| = \left|x-4\right|\left|x+4\right| < \tfrac{r}{9} \times 9 = r\,.$$